

IFB240 Risk Report

Samsung Galaxy S23 128GB



([Dr Debox, 2023](#)) ([Stick PNG, n.d.](#))

Student Table

Student ID	Student Name
n11400366	Lachlan Keates
n10773398	Rory Blair
n11048042	Diel Musgrove

Table of Content

<i>Introduction.....</i>	3
<i>Discussion.....</i>	4
<i>Context Establishment.....</i>	4
<i>Risk Assessment.....</i>	9
- Mobile Application Risk Report:.....	9
- <i>Mobile Device OS Risk Report:.....</i>	11
- User Behaviour Risk Report:.....	13
- Physical Threats Risk Report:.....	15
<i>Primary Impact Analysis.....</i>	17
- Privacy Policy Summary.....	17
<i>Risk Identification.....</i>	18
Risk Treatment and Countermeasures.....	19
Control Measures.....	19
- Mobile Application:.....	19
- Mobile Device OS:.....	19
- User Behavior:.....	20
- Physical Threat:.....	20
Recommendations.....	20
People:.....	20
- Control Measure 1:.....	20
Policy and Process:.....	20
- Control Measure 1:.....	20
Technology:.....	20
- Control Measure 1:.....	21
<i>References (APA - 7th Edition).....</i>	21
<i>Appendix.....</i>	24

Introduction

The client organisation is a large electricity supply company called Energen. Energen aims to supply people with affordable electricity plans for homes and supply This company issues company mobile phones for easy communication between employees. The manager of this company uses a mobile device given to by the organisation. They use a Samsung Galaxy S23 128GB. The manager requires this phone in order to communicate efficiently with their team. In this organisation, company phones are used mainly for work-related purposes with some limited personal use. The employees use their phones to send emails to co-workers and clients, as well as make calls and access the work portal in which all the employees can discuss in a group context. The main concern with the use of the company phones, however, is the connection to IoT devices. The phones are equipped with Bluetooth Multimeter apps that can connect to external devices. The manager's role at Energen is to direct staff, organise jobs and clients, and oversee the budget and standards of the jobs done by the company. The manager has access to all work-related apps that have been mentioned as well and is allowed to use the phone for limited personal use such as texts, calls, emails and limited browsing.

This report aims to highlight the threats, vulnerabilities, and security incidents that come with the use of the company phones at Energen. It will highlight how the confidentiality, integrity and availability can potentially be compromised and how this can be mitigated. This report will also highlight the risk treatments and countermeasures for the risks identified that can be taken in order to ensure maximum safety and security within the organisation and provide recommendations relating to the people, policy and processes, and technology. According to the University of North Dakota ([University of North Dakota., n.d.](#)), data leakage is one of the main concerns in corporations with mobile devices. This can come in the form of malware programs installed on IoT devices or workers installing seemingly harmless apps and granting permission to apps without knowing what they are granting. This can give attackers access to secure servers in the workplace. As the manager has access to IoT devices through their phone as well as the access to personal apps, this is a major concern and needs to be made aware by employees. This report will provide a risk assessment based on articles relating to relevant security incidents as well as a privacy analysis for an app installed on the device including the privacy policy and potential risks that come. (limitations and assumptions made)

Discussion

Context Establishment

Hardware:

The device that the client uses is a Samsung Galaxy S23 128GB. This is a device that is owned by Energen and used by the client for managerial purposes as well as limited personal use such as personal messages and calls which does not impede the employee's job performance.

The mobile device contains the following hardware and software assets:

- **Processor (CPU): Octa-Core Kryo Processor**
 - 1 x 3.36GHz Kryo Prime core
 - 4 x 2.8GHz Kryo Gold cores
 - 3 x 2.0GHz Kryo Silver cores
- **Display**
 - Size: 153.9mm (6.1" full rectangle)/ 149.9mm (5.9" rounded corners)
 - Resolution: 2340 x 1080 pixels (FHD+)
 - Refresh Rate: 120Hz
- **General Camera Information**
 - Capable of Auto Focus
 - Capable of Flash
 - Video Recording: UHD 8K (7680 x 4320) @30fps
 - Slow Motion: 960fps@FHD
- **Rear Camera**
 - Resolution (Multiple): 50.0MP, 10.0MP, 12.0MP
 - Zoom: 3 x Optical Zoom, 30 x Digital Zoom
- **Front Camera**
 - Resolution: 12.0MP
- **Memory/Storage**
 - Memory: 8GB
 - Storage: 128GB
- **Battery**
 - Li-Ion 3900 mAh, non-removable
- **Connectivity**
 - Bluetooth 5.3 connectivity
 - Wi-Fi: 802.11

Hardware information disclosed by Samsung ([Samsung, 2023](#)).

Software:

Due to the nature that the device is used, being a work-related device with only limited personal use, the device will be limited to certain applications for work related purposes. Without any personal software downloaded the software will be limited to only that required for work, this being:

- Android 13 with One UI 5.1
- Microsoft Teams (To communicate with employees)
- Zoom (To communicate with employees and clients)
- Dropbox (Sharing of files for projects)
- Microsoft Office (Productivity)
- SmartMeter (Work on site)

Data:

The data that is contained on this phone is mostly sensitive information to the company.

The data stored on this information would be as followed:

- Operating System: 30GB
- Documents: 8GB
- Photos/Videos: 8GB
- Emails (with attachments): 15GB
- Contacts: 1GB
- Other applications (listed above): 10GB

Overview of Device:

A Samsung Galaxy S23 128GB is loaned out by Energen to the client to perform managerial duties. This device is used primarily as a tool for conducting business but is also used for limited personal use such as calls, texts and internet browsing if it does not impede the work.

The data that is contained on this phone is mostly sensitive information for the company, due to that being the device's primary use. As a manager many important documents and records are stored on this device. This includes sensitive information pertaining to the employees beneath the management, information such as their name, email address, work schedule, emergency contacts, address, and date of birth. Information pertaining to the company's practices are also contained on this phone. Information such as expenses, orders, contracts with clients and client information will be stored on this device.

Most of this data would be considered confidential, such as the employee information and client information, making this data especially critical for the business.

Applications Used:

Many of the most used applications will be ones to communicate with the employees and clients as well as organise projects as a manager. Many of the applications which would be frequently used would be:

- Microsoft Teams:

This application will be used to communicate with other employees and general organisation. The main risk associated with this application would be from user error, from a situation such as a third-party attacker impersonating an employee and gaining access to the group, having confidential information exposed to an individual that should not have access to it ([Arimoto et al., 2023](#)).

One way to combat this risk, would be to ensure that you are only giving access to authorised users, and consistently checking that only authorised users are within the group, retaining the confidentiality of the project.

- Zoom:

This application is primarily used for video chat, with employees and clients. This application is similar to Microsoft Teams, where the major risk of using this application would be an unauthorised third-party joining a private call discussing confidential information ([Zoom, 2023](#)).

This risk can be mitigated by ensuring the calls are done privately, and the client is certain only authorised users are within the call, ensuring the confidentiality of the information.

- Dropbox:

Dropbox is a cloud-service that specialises in file-sharing between users. This will be used in Energen between employees to upload plans and other resources about the projects that are assigned, so they can be accessed by others working on the same project ([Johnson, 2021](#)).

Dropbox is a moderately secure application, with much of the risk being ones of human error. These risks can be avoided by taking care when uploading files, to ensure they are in the right spot, and encrypting particularly sensitive files before uploading them with a third-party application. Another aspect to take care of is to ensure permission to delete and modify files are not given to an unauthorised user. Taking precautions such as these helps maintain the confidentiality, integrity, availability in accordance to Energen's security goals.

- Microsoft Office

Though Microsoft Office is typically used for word and document processing, on this mobile device, it will mainly be used for its email feature, for contacting employees and clients ([Arimoto et al., 2023](#))

Being one of the largest productivity applications, Microsoft Office is quite secure in its file storing methods, with the user behaviour being the largest aspect at risk. To mitigate this risk ensure that you are aware what emails are legitimate or not, and to not give information or use any links or attachments from any emails perceived illegitimate.

- **SmartMeter:**

SmartMeter is an application that connects to an external peripheral, the Bluetooth multimeter, a device that is used to collect data on the project site ([Bush, 2013](#)) This device collects data such as voltage, current and resistance, then transmits this data from the device to the Samsung Galaxy S23 through Bluetooth.

SmartMeter's connection via Bluetooth leaves a vulnerability that a third-party attacker can exploit. This connection could lead to protected data being vulnerable to an attacker, leading to the compromise of Energen's security goals.

Risk Assessment

- *Mobile Application Risk Report:*

Title: Microsoft remedies three zero-days on February Patch Tuesday

Author: Tom Walat

Walat, T. (2023, February 15). *Microsoft remedies three zero-days on February Patch Tuesday*. TechTarget. Retrieved April 27, 2023, from: <https://www.techtarget.com/searchwindowsserver/news/365531374/Microsoft-remedies-three-zero-days-on-February-Patch-Tuesday>

Brief summary:

The article briefly mentions Microsoft fixing 76 total bugs in the February patch but goes into detail about a vulnerability in the Microsoft Office app on Android and IOS, which allowed for Remote Code Execution.

Information asset: The information that could be stolen from an attack described in the article is anything that is on the device in question. This includes not only files located on the device such as photos and documents, but could also obtain credentials and other valuable data located on the device.

Security issue:

Threat: The threat explained within the article is the threat of a Remote Code Execution, that could be performed by an attacker without needing any input from the user. A Remote Code Execution allows a malicious attacker to take control of a user's device and execute commands to do a myriad of attacks to the device.

Examples of attacks that could be performed by an RCE would be: (Checkpoint, n.d.)

- Information Disclosure
- Ransomware
- Denial of Service (DoS)
- Cryptomining

Vulnerability: The vulnerability presented within this article is simply having the vulnerable application (Microsoft Office) installed on a device. Having this application installed leads to the potential of an attack deemed critical by Microsoft.

Security incident/attack: Though the article does not refer to any specific attack that was made with this exploit, it discusses a security incident that is potentially highly severe if a malicious attacker were to capitalise on the vulnerability. An attack of this magnitude that allows an attacker to disrupt your data in a multitude of ways, clearly violating Energen's security goals.

With an RCE an attacker is able to write, view, modify and delete files, breaching all security goals of confidentiality, integrity and availability.

Energen's files being viewed, modified or removed have the possibility to be critical for the company. Vital information such as employee's emergency contact can be removed from the devices, leading to a physical safety risk for the employee. The modification or removal of data from the device would be detrimental to Energen.

- Mobile Device OS Risk Report:

Title: Own a Samsung phone? Install this update right away

Author: Prakhar Khanna

Khanna, P. (2023) *Own A Samsung phone? install this update right away*, *Digital Trends*. Digital Trends. Retrieved April 25, 2023, from:
<https://www.digitaltrends.com/mobile/samsung-galaxy-phone-april-2023-security-update-news/>

Samsung. (2023, April). *Security Updates Firmware Updates | Samsung Mobile Security*. Samsung Mobile Security. Retrieved April 25, 2023, from
<https://security.samsungmobile.com/securityUpdate.smsb#gnb>

Brief summary:

This article details security vulnerabilities found within the operating system of devices that were running Android 11 or above. The article explains that the issues were found by an internal security research team at Google, with them finding 55 issues, with many of them being considered critical or high priority to be fixed. The article also contains a list of devices affected and a link to Samsung's report detailing all the fixes contained in the update.

Information asset:

There were multiple types of data that were compromised and vulnerable before the update was released. This included information such as the location data of the device, as well as protected data within the device. These affected users of devices ran Android 11 or above and that were sold in Europe and Africa. Energen's data is highly sensitive, having confidential information such as employee's personal information, client information and project information.

Security issue:

Threat: This is an external threat that requires an outside attacker to prey upon the vulnerability. Prior to the update, an attacker was able to use unintended inputs in TIGERF Trustlet to access protected data. This attack could be caused by either accessing the memory location as it was at the end of its buffer or by using improper inputs the system was unable to register. Another threat that was fixed within the same update was another external threat that allowed an attacker to get access to the location of the device using the wifi network it was connected to. This was caused by an exploit that allowed an unauthorised user to control a vulnerability in SLocation.

Vulnerability: Two of the major vulnerabilities that were discussed in the article were the vulnerabilities of a user's data and the location of the user's device. Both exploits were caused by an attacker external providing inputs into software on the operating system to be able to view protected data such as the user's location and personal files.

Security incident/attack: The above article does not go into detail about a specific attack, but rather vulnerabilities pertaining to an operating system that affects multiple devices. This violates the confidentiality aspect of the security goals, as the data that can be accessed through this exploit was able to be viewed by a third party but cannot be modified or deleted in any way. Though this article only described a vulnerability within the operating system, and not an attack, this vulnerability affected millions of users of Samsung devices. Access to Energen's company data is critical to the conduct of their business, and has great amounts of sensitive data, pertaining to both customer and employee information.

- User Behaviour Risk Report:

Title: The Samsung Galaxy S23 series doesn't support Seamless Updates

Author: Hadlee Simons

Simons, H. (2023, February 6th). *The Samsung Galaxy S23 series doesn't support Seamless Updates*. Android Authority. Retrieved April 26, 2023, from <https://www.androidauthority.com/samsung-galaxy-s23-series-seamless-updates-3276559/>

Brief summary:

The above mentioned article discusses the lack of a seamless update system for the Galaxy S23 and also discusses the advantages of using seamless updates. The reason why seamless update systems are so widely used and commended is seen in the Android Open Source Project Docs. This document states that seamless updates ensure a workable booting system stays on the device while an OTA update is in process. It also reduces the possibility of an inoperative device after an update, which means less replacements and reflashes are needed ([Android Open Source Project, 2022](#)). The non use of this seamless update system can lead to the whole device being unusable or require a reset to factory defaults to be performed by the user.

Information asset:

The information assets that are at risk in this instance is the entire collection of data stored on the device, which could contain company and personal assets. Company assets could include company credentials and/or company/client details. Personal assets could include the storage of personal details (banking or medical credentials). This could be unfortunate if the only copy for both company assets and personal assets were secured on this one device at the time an update is performed. There are also two factor authentication applications that are often used on mobile devices. The loss of access to these can be at risk.

Security issue:

Threat: This would be considered to be an internal threat as the user must interact (turn phone off and then on again) to initiate the update process. There is no automatic backup performed. This could be a threat to the availability of confidential information assets as the problem (no A/B updates) could render the device useless and the user is unable to recover said information assets.

Vulnerability: The vulnerability is that the user may not always backup their phone to another device/cloud and may believe the device is creating a backup or the new update is not destroying the data retained on the device. The current update process, triggered by the turning on of the device does not prompt the user to make a backup prior to the update being performed. The use of seamless updates would remove this vulnerability from the S23.

Security incident / attack: Although no security attack/incident is mentioned within this article, with doing extra research about seamless updates and bloatware, multiple scenarios can be envisioned. One scenario as mentioned above, is that the user doesn't backup the device or thinks it will be backed up for them, when it is not. (**Vulnerability**). Then they initiate the update process without knowing the possible effect that could result (**Threat**). If the update has a flaw, which can occur, the device could be rendered inoperable (**Security Incident/Attack**). This scenario, if played out, would be considered a Security Incident as the user (Manager) had no intent to compromise the availability of information assets.

- Physical Threats Risk Report:

Title: Does the Galaxy S23 Have Face ID? All the Ways you can Unlock the Phone

Author: Kishalaya Kundu(Top/Face ID Article), Kanav Sood(Bottom/Fingerprint Article)

Kundu, K. (2023, February 12). *Does The Galaxy S23 Have Face ID? All The Ways You Can Unlock The Phone*. Screenrant. Retrieved April 26, 2023, from <https://screenrant.com/galaxy-s23-face-unlock-options/>

Sood, K. (2023, March 27). *Samsung Galaxy S23 fingerprint scanner or biometric unlock inconsistent or slow for some users (potential workarounds)*. PiunikaWeb. Retrieved April 27, 2023, from <https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

Brief summary:

The first article listed above explains how to set up facial recognition, the app/processes that require this biometric function to be used and the disadvantages of the Galaxy S23 face ID system compared to Apple's face ID system. These disadvantages could be used by attackers/thieves to gain access to the user's phone via an incorrect face ID match and use of this incorrect match could result in access to a limited number of other apps/systems. The S23 allows the accuracy required to use the face recognition app to be reduced by adjusting settings about the eyes being open and/or adjusting for faster recognition. (Verizon, 2023). The second article talks about stories of multiple users and their concerning interaction with the fingerprint scanner on the Galaxy S23. The main concern is about the use of the fingerprint scanner and the addition of a protective screen. Cold and heat can also play a part in the inconsistencies with the fingerprint scanner. (All Science Fair Projects, n.d.) (Clark, n.d.). This problem can be fixed in the fingerprint settings via adjusting the sensitivity level of the fingerprint function or removing the screen protector. Reducing the accuracy required by either security function puts the device and all information contained on it at risk.

Information asset:

The information assets that could be at risk via these two physical threats is some, if not all, data stored on the device. The first biometric feature, The Face ID, can get you past the lock screen and access to some functionality of the device. The second feature, the Fingerprint Scanner, is more serious. Whilst the Face ID is used to gain access to the home screen, it cannot be used for banking/wallet apps. These apps require the use of the Fingerprint scanner.

Security issue:

Threat: This would be an external threat as the attacker has a malicious intention of using either the user's face and/or the user's fingerprint via replicas created and/or collected. However, to do so the attacker must also have the user's phone in hand. This would initially be considered a threat to availability. However, after access is gained by the attacker, the integrity and confidentiality of information assets could also be compromised.

Vulnerability: The vulnerability is that the user doesn't have a way of tracking or locking out the attacker should they relax these biometric features. The removal of a screen protector so as to ensure consistency with the use of the fingerprint function also increases the risk of physical damage to the device. Physical damage to the device can result in loss of access to all data stored on the device.

Security incident/attack: Although the 2 articles don't mention any security incident/attacks, both forms of biometric security manipulation can be considered security risks. Should the phone be stolen and the user had altered/relaxed the settings in these biometric security settings, the attacker now could gain access to the phone and data thereon. (Threat). It may take the attacker several attempts to gain access to the phone but once they do the settings could be changed so that it is as if they are the device's owner. This then may allow them to use wallet apps (Security Incident/Attack). If this security attack scenario was to play out in the real world, it could allow the attacker to access additional personal information (Confidentiality), whilst the user cannot access or stop the attacker (Availability). With today's digital ID usage including digital drivers licenses, MyGov ID and other card storage capabilities, there may be an increased risk identity theft.

Primary Impact Analysis

- Privacy Policy Summary

The Privacy Policy for Zoom and all the information regarding privacy and security can be accessed from the 'Zoom Privacy Statement' as well as the 'Privacy & Security for Zoom Video Communications' pages from Zoom's website.

The type of data that Zoom collects is Personal data. This is data such as account details that have been given by the user, the details of the user's device and content that has been shared in meetings. Zoom uses tools that can automatically detect the content and store it. Employees may also directly access the content if they are required to for the sake of legality or safety or if given explicit permission from a user. Zoom can access this information at any time, if the information has not been deleted.

The information that is received by Zoom is mostly relevant to the user's purpose. They request, with permission from the user; account information such as names and date of birth, contact information such as phone numbers and emails, transactional information like card numbers and details. They also have access to the device make and model, specifications, and settings within the app to be able to run the features optimally. However, Zoom also is allowed to keep content from the collaborative features like meetings and chats. This is content such as audio, video, messages from chat, and answers to polls. Finally, Zoom has permission to access your information from third party apps including market data, to better target advertisements to you. Zoom states that the information obtained by their service includes: To provide Zoom products and services, Product research and development, marketing promotions, to communicate to the users, and to maintain authentication, integrity, security and safety.

For file transfer storage, especially, this data is stored in the cloud, encrypted, and deleted after 31 days. Whether the data is being transmitted or stored, Zoom encrypts it using AES 256 which is a symmetric encryption method that uses a one-time key. If the information is only stored locally, the user has the option to encrypt it, but it will not automatically be encrypted. Zoom is allowed to share any personal information with Vendors, Resellers and Third-Party developers. If required content is also shared if it is needed for legal reasons with law enforcement and government agencies. The app user is allowed to access their own personal data at any time and modify it when needed. They have access to recordings, product usage, participant list, registration information, and the feedback to polls made by the user. The user can access this information, at any time, through the app.

Risk Identification

Zero Click Attack

A newly discovered Zoom bug poses a major risk to the company of Energen. A type of Zero Click Attack was found by Google's Project Zero Team ([HI TECH, 2022](#)). A Zero Click Attack is a type of bug with malware that does not need to be accessed by the victim and will be successful if it is sent to the victim. This attack compromises confidentiality as the malware is usually in the form of spyware. This can be used to access information that the hacker has not been granted with. The impact is quite severe for Energen as hackers could have access to sensitive information. The threat of this situation is the bug itself containing the malware. The vulnerability comes from the fact that Zoom allows these files to be shared.

Low-Standard Encryption:

Zoom meetings and conversations can contain very important and confidential information for companies and organisations. It is very important that messages that contain this data are heavily encrypted. According to Sonia Imam of PureVPN ([Imam, 2023](#)), despite claims from Zoom that they offered up to standard encryption, it was found that it had not been the case. Hackers were able to bypass the low-level encryption and intercept data. This directly compromises confidentiality that is to be assumed when using this app. The manager of Energen would likely be using Zoom to discuss private information with the team. By doing this, Zoom has a major vulnerability in the system which would be the low-level of encryption being used to transmit data. The threat would be malicious by intended people looking to exploit it. The impact that this could have would be catastrophic to Energen private information as they would not like the details of employees and clients leaked.

Easily hackable meeting IDs via 'zWarDial':

The information assets that are most at risk is the information discussed during the meeting as well as the host's name. The vulnerability is that the Zoom has a low-standard encryption, possibly allowing the threat to act upon it. The threat is that the attacker is able to use a tool, called 'zWarDial', to get access to details about a particular meeting. If a security incident was to happen within the company, the device would not be affected, but the confidentiality of the meeting may affect the manager and the organisation. The risk level for an event like this would be high to medium as only the confidentiality is compromised, depending on the topic, but not availability and integrity. It is still available to the manager and the organisation and the attacker can only view information about the meeting, not change it.

Risk Treatment and Countermeasures

Control Measures

- *Mobile Application:*

Overview:

The article briefly mentions a vulnerability found within the Microsoft Office application for Android, allowing for a potential remote code execution. This could potentially lead to an unauthorised user gaining access to potentially confidential information such as client information and trade secrets. The article does not relay any specific attack or any data reflecting the incident, however it can be inferred that this vulnerability would affect a large number of people, as Microsoft Office is one of the leading email applications on the Android operating system.

Treating the Risk:

Suggested Control Measure: The vulnerability discussed within this application is one that was considered to be of maximum severity, with its confidentiality, integrity and availability all being rated as 'high' in terms of how much they are affected (Microsoft, 2023). With the exploit not requiring any input from the user to be accessible, the best control measure for this situation would be to uninstall the application and use another for emails.

Control Measure Type: The type of control measure suggested is a preventative measure, one that would completely eliminate the vulnerability though at the cost of an inconvenience to the user, where they would have to setup and begin using another application for their emails

Level of Protection Provided: The level of protection that is granted through this control measure is one that is capable of eliminating all risk associated with using the application, as without using the application, an attacker will have no way of exploiting the application's vulnerability.

Control Measurement Limitations: The limitations of this control measurement are that despite it eliminating all risk, it would be troublesome to change all employee's email applications when Energen uses the same application for all employees. However, this would be worth the effort as the consequences of not taking any action to prevent this risk would be dire.

- Mobile Device OS:

Overview:

The article briefly mentions a security vulnerability that was present within devices that were running Android 11 or below. There were 55 vulnerabilities found within this by an internal security team at Google. Many of the vulnerabilities found within the operating system were deemed critical. Information such as the user's location as well as any data that was stored on the phone.

Treating the Risk:

Suggested Control Measure: A control measure for this scenario would be to avoid connecting to networks that are public or are unsure of their safety. As Energen requires the user to go to different sites, they may need to connect to different networks, some of which are not as secure as others. As the exploit with the operating systems preys on the user's connection to an external network, this risk can be mitigated by ensuring only trusted networks are connected to.

Control Measure Type: The type of control measure being suggested is a preventative measure, making less opportunities for the vulnerabilities to occur throughout the use of this device. This method will not be able to stop all potential attacks as connecting to an external network is required for Energen's work.

Level of Protection Provided: The level of protection that would be applied through this method would not eliminate the risk associated with the operating system's vulnerability. You would not be able to eliminate connection to the internet, as that is essential for the work requirements at Energen. The protection granted by this suggested method relies upon user's own judgement and abilities to recognise when it would be safer to connect to certain networks and when to not

Control Measurement Limitations: As mentioned previously, this method does not remove all risk of using the device, as when using the device and connected to an external network the user and their data is vulnerable. Another limitation would be that the user is required to make their own judgment about what networks are safe to connect to and to assess whether taking the risk of potentially having a vulnerability exploited.

- User Behavior:

Overview:

As discussed previously, a vulnerability with regards to the seamless updates was discussed. This vulnerability could, in the event of a fault in an installed update, render the Galaxy S23 device unusable. This security incident could occur when an update is initiated by the user by simply turning the device off and on again. The use of seamless updates always makes sure a workable booting system is left on the device during an update process, and this is currently not the case with the Galaxy S23. (Simons, 2023). This security incident could impact the availability of info assets that were stored on the device, thus availability of data and information is lost.

Treating the Risk:

Suggested Control Measure: A control measure for this risk would be the use of the Samsung Cloud, a backup and cloud storage service offered by Samsung. This control measure has 2 ways of operating, manual or automatic. Manual operation requires the user to go into the settings and press a button for the backup to initiate. The Automatic operation requires the user to turn a setting on, which will then backup the device once every 24 hours (Galaxy S23 Manual, n.d.). Both of these options can be used as they complete the same objective.

Control Measure Type: The use of automatic backup would be a preventive control measure as this control measure is preventing the loss of information assets before they are put at risk during the update process. This control measure is not a detective control measure as it is not monitoring any incidents nor is it a corrective control measure as it is not detecting and repairing any damage or loss of information assets. This countermeasure is effective if used properly in everyday use as any update could create the security incident.

Level of Protection Provided: This Control Measure has a high level of protection, as it is a countermeasure to the vulnerability, but also has its flaws. For instance, if an auto sync backup took place and then later the security incident was to occur, the information assets on the backup would not include any changes made between that auto sync backup and the time of the incident. If the manual backup procedure was implemented and the operator did not perform the backup prior to initiating the update, there is a risk of a larger volume of information assets being lost or damaged.

Control Measurement Limitations: One limitation of this control measure is the way the backup is performed. If the user is using the manual backup option, then it would be up to the user to backup the device before initiating the update. Another limitation of this control measure is that if someone were to get personal (confidential) information assets, such as email and password for the cloud service, then the intruder is able to access all the information assets in the Samsung Cloud.

- Physical Threat:

Overview:

The risk assessments undertaken above, uncovered a physical vulnerability that could be used for malicious intent. This vulnerability concerns the inconsistencies of security measures, like face recognition and the fingerprint scanner, physical biometrics. These biometrics could be used to get past the lock screen, access certain settings and in some cases access and use of a banking application, possibly causing damage to the information assets on the device. This security incident, or in some case security attack, will affect all security goals (Confidentiality, Integrity and Accountability).

Treating the Risk:

Suggested Control Measure: The control measures that would be suggested to counteract this vulnerability is to ensure all security settings are applied and certain settings are set as required to achieve the required level of security desired. This is done by ensuring the faster recognition option for face ID is turned off and that the sensitivity for fingerprint recognition remains set on high. This would help to ensure that biometric security can not be used to change settings or access confidential information assets, and running security updates.

Control Measure Type: The control measures being suggested would be preventive and detective. Preventive, as these controls should prevent the unauthorized access to particular information assets. Detective, to the extent that they detect incorrect access attempts and then block access to the device and information stored thereon. They are detective to the extent that they are always monitoring as in being ready to detect access attempts.

Level of Protection Provided: These control measures would provide a moderate level of protection as they will provide the bare minimum in security when on factory default. With some extra features turned on, the level of protection would be a high to moderate level, however, it still does not provide the best security options. This can be linked to the fact that with the best security features active on the device, some features, such as fingerprint and face ID, can still be inconsistent and lead to a security attack as a result of issues with the camera software.

Control Measurement Limitations: The limitations of these control measures is that even with the high level security options active, the device can still be accessed by unauthorized users. Control measures will never eliminate the vulnerability and threat but only minimize the likelihood of the threats acting upon the vulnerability, thus a security incident/attack taking place. These control measures are subject to some risk in that they also require a reliance on the operator not to relax any settings.

Recommendations

People:

People are often the cause of data breaches or security incidents and it is usually completely unintentional. This can be achieved in many ways, most prominently however, are phishing attacks. A study conducted by Stanford University in 2020, confirmed that 88% of all data breaches are caused by an employee making a mistake and states that human error is still one of the main concerns to cyber security.

We recommend that all employees at Energen undergo a mandatory security awareness training course in order for employees to understand the risks involved and to use safe practices when dealing with online services and important information. Sentrion is a Australian and New Zealand based company that offers a 15 minute course and is rated 4.7 stars on Google (Sentrion, n.d.). As well as training, employers can adopt cyber safe practices that everyone must follow. This means that Energen should provide a list of exactly what can and cannot be done in regards to using the internet.

By introducing this, Energen can ensure that the confidential information that employees have access to are not accidentally leaked and minimal data breaches should occur. Proper education and safe practices are essential in any business dealing with the internet.

Policy and Process:

For Energen, or any company, policies and processes must be enforced in order to maintain the security of the business. Business.gov.au has a guideline on how to create a good cyber security policy that can be implemented (Australian Government, 2023).

The policy should include password requirements that ensure all employees are using strong, unique passwords. It should also highlight how to use email safely and securely. This means knowing when and where a work email can be shared, and making sure junk and spam are blocked. Another inclusion should be explaining how to handle sensitive data and setting rules around how technology is used to ensure this. Employees should be aware that they are using sensitive data and know who and where they can share this data. They should also take physical precautions like not using the internet on an unsecure network, storing the device in a safe location, and make sure they are only using trusted devices such as flash drives.

Ensuring that a policy is in place and that employees are following specific processes can help mitigate the human error that causes data breaches and make the information of the company secure.

Technology:

We recommend that Energen implement a few changes to the way employees use the technology. As stated previously, a security policy should be introduced and followed by all personnel. Changes to the way we use technology can drastically help the security and mitigate data breaches.

All employees should use strong, hard-to-guess passwords for all devices, and programs and have vary. It is very problematic if an attacker is able to identify someone's password, especially if they use that same password for everything else (Get Cyber Safe, 2021). As well as a strong and unique password for all apps and devices, Energen should ensure that these passwords are to be changed frequently. It is important, also, that these passwords aren't similar to the original password. This is because if someone has access to a previous password, they can easily guess what the new one could be.

Ensuring that all software and operating systems are always kept up to date is essential. Developers fix bugs and vulnerabilities that have been exposed in each update. If software is not kept up to date, then attackers are able to exploit these known vulnerabilities and violate the integrity, availability, and confidentiality of the data.

Finally, it is recommended that antivirus software is installed and kept up to date. This will prevent any suspicious activity that could potentially cause a major security incident.

References (APA - 7th Edition)

(This is the Reference Format)

Author. (Publish Year, 'Publish Month' 'Publish Day'). *Title*. Website Name. Retrieved Access Month Access Day, Access Year, from URL

All Science Fair Projects. (n.d.). *The Effect of Temperature on Fingerprints*. All Science Fair Project. Retrieved April 28, 2023, from <https://www.all-science-fair-projects.com/project1154.html>

Android Open Source Project. (2022, September 13). *A/B (Seamless) System Updates*. Android Open Source Project. Retrieved April 26, 2023, from <https://source.android.com/docs/core/ota/ab>

Arimoto, P., Rakshith, P., Brunn, G., Mathew, J., Dharmanayagam, B., Cole, L. H., Kjerland, S., & Coulter, D. (2023, January 26). *Microsoft Teams service description*. Microsoft. Retrieved April 26, 2023, from <https://learn.microsoft.com/en-us/office365/servicedescriptions/teams-service-description>

Arimoto, P., Rakshith, P., Dharmanayagam, B., Cole, L. H., Coulter, D., Kjerland, S., Smith, S., Vangel, D., & Baumgartner, P. (2023, January 26). *Office applications details*. Microsoft. Retrieved April 26, 2023, from <https://learn.microsoft.com/en-us/office365/servicedescriptions/office-applications-service-description/office-applications>

Bush, S. (2013, September 4). *Bluetooth multimeter uses phone as display*. Electronics Weekly. Retrieved April 26, 2023, from <https://www.electronicsworld.com/blogs/engineer-in-wonderland/general-engineer-in-wonderland/bluetooth-multimeter-uses-phone-as-display-2013-09/#:~:text=Bluetooth%20Multimeter%20is%20an%20intelligent,communicate%20with%20phones%20via%20bluetooth>

Checkpoint. (n.d.). *What is Remote Code Execution (RCE)?* Checkpoint. Retrieved April 26, 2023, from <https://www.checkpoint.com/cyber-hub/cyber-security/what-is-remote-code-execution-rce/>

Clark, M. (n.d.). *False Acceptance in Winter: Why Do Fingerprint Scanners Fail in Cold Weather?* Bayometric. Retrieved April 28, 2023, from <https://www.bayometric.com/false-acceptance-in-winter-why-do-fingerprint-scanners-fail-in-cold-weather/#:~:text=There%20are%20many%20things%20that,false%20acceptance%20and%20false%20rejections>.

- Dr Debox. (2023, March 3). *Samsung Galaxy S23 Unboxing - ASMR*. Youtube. Retrieved April 26, 2023, from <https://www.youtube.com/watch?v=yfcypVwFz7M>
- Galaxy S23 Manual. (n.d.). *How to Back Up Galaxy S23 to Samsung Cloud Automatically*. Galaxy S23 Manual. Retrieved 06 01, 2023, from <https://www.galaxys23manual.com/back-up-galaxy-s23-to-samsung-cloud/>
- HT TECH. (2022, August 22). *Zoom ALERT! iPhone, Android or Windows users are at HIGH risk; Do this now*. HT TECH. Retrieved April 26, 2023, from <https://tech.hindustantimes.com/tech/news/zoom-alert-iphone-android-or-window-s-users-are-at-high-risk-do-this-now-71653630882035.html>
- Imam, S. (2023, March 17). *Zoom security issues: How to stay protected on flaw-riddled Zoom*. PureVPN. Retrieved April 28, 2023, from https://www.purevpn.com/blog/zoom-security-issues/#What_are_Zooms_security_issues
- Johnson, D. (2021, March 5). *Is Dropbox secure? Here's how Dropbox has improved its security measures, and what you can do to protect yourself*. Insider. Retrieved April 26, 2023, from <https://www.businessinsider.com/guides/tech/is-dropbox-secure#:~:text=Because%20Dropbox%20is%20relatively%20secure,consider%20using%20file%2Dlevel%20encryption>
- Khanna, P. (2023, April 5). *Own a Samsung phone? Install this update right away*. DigitalTrends. Retrieved April 25, 2023, from <https://www.digitaltrends.com/mobile/samsung-galaxy-phone-april-2023-security-update-news/>
- Kundu, K. (2023, February 12). *Does The Galaxy S23 Have Face ID? All The Ways You Can Unlock The Phone*. ScreenRant. Retrieved April 26, 2023, from <https://screenrant.com/galaxy-s23-face-unlock-options/>
- Microsoft. (2023, February 14). *Windows Graphics Component Remote Code Execution Vulnerability*. Microsoft. Retrieved 05 31, 2023, from <https://msrc.microsoft.com/update-guide/vulnerability/CVE-2023-21823>
- Samsung. (2023, February 21). *Specifications Galaxy S23 128GB*. Samsung. Retrieved April 26, 2023, from <https://www.samsung.com/au/smartphones/galaxy-s23/specs/>
- Samsung. (2023, April). *Security Updates Firmware Updates*. Samsung Mobile Security. Retrieved April 25, 2023, from <https://security.samsungmobile.com/securityUpdate.smsb#gnb>

- Sansom, A. (2023, February 07). *Samsung Galaxy S23 hides a nasty storage secret*. Tom's Guide. Retrieved April 27, 2023, from <https://www.tomsguide.com/news/samsung-galaxy-s23-hides-a-nasty-storage-secret>
- Simons, H. (2023, February 6). *The Samsung Galaxy S23 series doesn't support Seamless Updates*. Android Authority. Retrieved April 26, 2023, from <https://www.androidauthority.com/samsung-galaxy-s23-series-seamless-updates-3276559/>
- Sood, K. (2023, March 27). *Samsung Galaxy S23 fingerprint scanner or biometric unlock inconsistent or slow for some users (potential workarounds)*. PiunikaWeb. Retrieved April 27, 2023, from <https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>
- Stick PNG. (n.d.). *Download Warning Icon transparent PNG*. Stick PNG. Retrieved April 26, 2023, from <https://www.stickpng.com/img/miscellaneous/safety-symbols-and-signs/warning-icon>
- University of North Dakota. (n.d.). *Mobile Device Security Risks for Corporations*. University of North Dakota. Retrieved April 26, 2023, from <https://onlinedegrees.und.edu/blog/mobile-device-security-risks-for-corporations/#:~:text=Data%20Leakage,then%20access%20secure%20servers%20undetected>
- Verizon. (2023). *Samsung Galaxy S23 / Galaxy S23 Ultra - Set Up Face Recognition*. Verizon. Retrieved April 26, 2023, from <https://www.verizon.com/support/knowledge-base-304078/#:~:text=Samsung%20Galaxy%20S23%20%2F%20Galaxy%20S23%20Ultra%20%2D%20Set%20Up%20Face%20Recognition,-2023%2D04%2D26&text=Facial%20recognition%20is%20utilized%20to,or%20away%20from%20direct%20sunlight>.
- Walat, T. (2023, February 15). *Microsoft remedies three zero-days on February Patch Tuesday*. TechTarget. Retrieved April 27, 2023, from <https://www.techtarget.com/searchwindowsserver/news/365531374/Microsoft-remedies-three-zero-days-on-February-Patch-Tuesday>
- Zoom. (2019). *Privacy & Security for Zoom Video Communications*. Zoom. Retrieved April 26, 2023, from <https://explore.zoom.us/docs/ent/privacy-and-security.html>
- Zoom. (2023, March 20). *Zoom Services Description*. Zoom. Retrieved April 26, 2023, from <https://explore.zoom.us/en/services-description/>

Zoom. (2023, April 10). *Zoom Privacy Statement*. Zoom. Retrieved April 26, 2023, from <https://explore.zoom.us/en/privacy/>

Appendix

Appendix 1 - Mobile Application Risk Report Article

Microsoft remedies three zero-days on February Patch Tuesday

The company releases its security updates to shut down the vulnerabilities threat actors have been actively using for attacks in the wild.



By Tom Whitel, Site Editor

Published: 16 Feb 2022

Microsoft addressed 76 security updates for February Patch Tuesday, including three zero-days.

There were no revised or updated vulnerabilities this month. In total, Microsoft patched nine CVEs rated critical and 66 rated important. One CVE (CVE-2019-15126) issued by the Mitre Corporation details how to stop exploits of HoloLens 1 devices, which uses unsupported Broadcom hardware.

Microsoft resolves three zero-days

Microsoft fixed a zero-day (CVE-2023-21823) rated important in [Universal Windows Platform apps](#) and Windows desktop and server systems. The title of the CVE indicates this is a Windows Graphics Component vulnerability, but it also affects Microsoft Office app on Android and iOS devices. Organizations that still run [Windows Server 2008/2008 R2 workloads in the Azure cloud platform](#) as part of the Extended Support Update program will also receive a fix for this bug.

No user interaction is required to exploit the flaw. It gives the attacker system privileges for

complete control Windows OS systems or the ability to perform a remote-code execution on the Android and iOS devices.

The second zero-day (CVE-2023-21715) is a Microsoft Publisher security features bypass vulnerability rated important for Microsoft 365 Apps on 32-bit and 64-bit systems. By default, these [Click-to-Run](#) applications update automatically by pulling the latest version from a network location.

Ivanti's vice president of security product management Chris Goettl said this flaw requires user interaction and could be easily done by emailing a link to a web site that hosts a specially crafted file and enticing the user to open it.

Chris Goettl "Phishing is not very difficult. It's more of a statistical game to get an authenticated user on the targeted system to open that file," Goettl said.

After the exploit succeeds, the threat actor can [bypass Microsoft Office macro policies](#) used to block untrusted or malicious files to avoid detection and move unchecked across the organization's network.

The last zero-day is a Windows common log file system driver elevation-of-privilege vulnerability (CVE-2023-23376) rated important for Windows desktop and server systems. Exploitation of this flaw does not require user interaction and would give the attack system privileges.

This CVE and CVE-2023-21823 are examples of the types of vulnerabilities a threat actor would use as part of their breach arsenal.

"Neither of them are dangerous on their own. But they would be used in combination with some

Sponsored News

[Modernize Your Microsoft SQL Server-Based Apps With a Flexible, As-A-Service ...](#)
— Dell Technologies

[Partners Take On a Growing Threat to IT Security](#)
— IBM

[Why Zero Trust, Why Now](#)

[Why Zero Trust, Why Now](#)
— Microsoft

[See More](#)

Related Content

[Microsoft fixes three zero-days in February update](#)
— ComputerWeekly.com

[Microsoft mends Windows zero-day on April Patch](#)
— SearchWindowsServer

[Microsoft resolves Windows zero-day on January Patch](#)
— SearchWindowsServer

other attack," Goettl said. "One could be used to get onto the box to elevate their privileges. Then from there they could play around with admin tools to become very hard to detect."

Exchange Server bugs eliminated

Microsoft repaired four remote-code execution flaws (CVE-2023-21529, CVE-2023-21706, CVE-2023-21707 and CVE-2023-21710), all rated important, in Exchange Server this month. The on-premises mail-server system remains a [highly valuable target for hackers](#) who, if they manage to exploit the system, will be free to move laterally throughout an organization's infrastructure.

"From what we've seen the last couple of years with the level of sophisticated attacks on Exchange, I would urge organizations to get these updates done in a timely fashion," Goettl said. "I would say you don't want to let this hang out there for longer than a month."

Multiple vulnerabilities hit SQL Server

Microsoft released nine patches to correct flaws related to its SQL database product on February Patch Tuesday:

- CVE-2023-21704: Microsoft ODBC Driver for SQL Server remote-code execution vulnerability, rated important;
- CVE-2023-21718: Microsoft SQL ODBC Driver remote-code execution vulnerability, rated critical;
- CVE-2023-21568: Microsoft SQL Server Integration Service (VS extension) remote-code execution vulnerability, rated important;
- CVE-2023-21528: Microsoft SQL Server remote-code execution vulnerability, rated important;
- CVE-2023-21705: Microsoft SQL Server remote-code execution vulnerability, rated important;
- CVE-2023-21713: Microsoft SQL Server remote-code execution vulnerability, rated important;

- CVE-2023-21685: Microsoft WDAC OLE DB provider for SQL Server remote-code execution vulnerability, rated important;
- CVE-2023-21686: Microsoft WDAC OLE DB provider for SQL Server remote-code execution vulnerability, rated important; and
- CVE-2023-21799: Microsoft WDAC OLE DB provider for SQL Server remote-code execution vulnerability, rated important.

Goettl said this was the most SQL-related security fixes in a single Patch Tuesday release since 2009. The number of patches paired with the complexity of a typical SQL Server environment will require coordination between the IT people who patch the operating system, the application owners and the SQL database administrator for a [successful patch deployment](#).

"There could be challenges with homegrown apps that can be sensitive to updates," Goettl said. "Usually, they need some testing to confirm things are good before they put it back in production."

Critical Microsoft Word vulnerability requires prompt attention

Administrators will also want to put a Microsoft Word remote-code execution vulnerability (CVE-2023-21716) rated critical at the top of their [patching priority](#) list. The preview pane in Microsoft Outlook is an attack vector, which means the recipient only needs to view the malicious email to give the threat actor access to their system.

Microsoft lists this bug with a CVSS rating of 9.8 out of 10.

"It's definitely going to be big threat if somebody figures out how to start taking advantage of it," Goettl said.

Appendix 2 - Mobile OS Risk Report Article 1

Own a Samsung phone? Install this update right away



By Prakhar Khanna
April 5, 2023

SHARE

Samsung has begun rolling out its latest security update, and it's crucial for those with phones using Exynos modems — like last year's [Galaxy S22](#) flagship — which were at high risk of major security breaches. The new update fixes these security concerns (and more) and is now rolling out to the [Galaxy S23](#) lineup in Europe and South Korea.

In mid March, a Project Zero report from Google's internal security research team revealed that [security issues found on devices using Exynos modems](#) could have given remote users the ability to very easily compromise a phone at the baseband level. Samsung said at that time that it released security patches for five of those vulnerabilities in March, with an update to follow in April to address the remaining issues.

According to a report published on [Samsung's website](#), the April 2023 security patch update contains fixes (CVEs) from Google for 55 security flaws, out of which four fixes are termed critical and 46 fixes are termed high priority. That is why it is important to install this update on your Galaxy phones as soon as it arrives, despite the [Galaxy S23](#) series not using the Exynos modem. It also fixes the security vulnerabilities found in 3GPP implementation in the Exynos modem baseband and Exynos fastboot USB interface.

Samsung is also rolling out 15 additional fixes with the update for Galaxy smartphones and tablets. Most of these flaws were found on devices running [Android 11](#) and above. Apart from the high-risk vulnerabilities, flaws in SemClipboard, SLocatoin, and Telephony features have also been fixed alongside taking care of loopholes found in the TIGERF trustlet, hijacking vulnerability in CertificatePolicy as well as in improper authorization of the Smart Suggestions widget. In other words, a *lot* of vulnerabilities have been patched.

Galaxy devices, including the S22 sold in Europe and African countries, M33, M13, M12, A71, A53, A33, A21s, A13, A12, and A04 series were at risk according to the Project Zero list — we expect them to receive this security update soon. All eyes are now on Google to roll out this crucial fix for its Pixel 6, [Pixel 6a](#), Pixel 6 Pro, [Pixel 7](#), and [Pixel 7 Pro](#).

Appendix 3 - Mobile OS Risk Report Article 2

SVE-2022-2948(CVE-2023-21470): Improper access control vulnerability in SLocation

Severity: Moderate

Affected versions: Android 11, 12, 13

Reported on: December 13, 2022

Disclosure status: Privately disclosed

Improper access control vulnerability in SLocation prior to SMR Apr-2023 Release 1 allows local attackers to get device location information using com.samsung.android.wifi.NETWORK_LOCATION action.

The patch change to explicit intent.

SVE-2022-2318(CVE-2023-21478): Improper input validation vulnerability in TIGERF trustlet

Severity: High

Affected versions: Android 11, 12, 13

Reported on: September 19, 2022

Disclosure status: Privately disclosed

Improper input validation vulnerability in TIGERF trustlet prior to SMR Apr-2023 Release 1 allows local attackers to access protected data.

The patch add proper input validation.

SVE-2022-2315(CVE-2023-21477): Access of Memory Location After End of Buffer vulnerability in TIGERF trustlet

Severity: Critical

Affected versions: Android 11, 12, 13

Reported on: September 19, 2022

Disclosure status: Privately disclosed

Access of Memory Location After End of Buffer vulnerability in TIGERF trustlet prior to SMR Apr-2023 Release 1 allows local attackers to access protected data.

The patch add proper input validation.

Appendix 4 - User Behaviour Risk Report Article

28/04/2023, 18:35

The Samsung Galaxy S23 series doesn't support Seamless Updates

Q

ANDROID AUTHORITY

≡

🛒 Pixel 7: Just \$450 - Limited Time Deal!

Affiliate links on Android Authority may earn us a commission. Learn more.

MOBILE

The Samsung Galaxy S23 series doesn't support Seamless Updates

It looks like Samsung is sticking to the old method of installing updates.

By Hadlee Simons • February 6, 2023



Ryan Haines / Android Authority

<https://www.androidauthority.com/samsung-galaxy-s23-series-seamless-updates-3276559/>

1/5

28/04/2023, 18:35

The Samsung Galaxy S23 series doesn't support Seamless Updates

The Samsung Galaxy S23 series reportedly lacks support for Seamless Updates.

This is a more convenient method for installing system updates compared to the conventional process.

Samsung is one of several smartphone brands that haven't implemented **Seamless Updates** (also known as A/B updates) on their phones. Unfortunately, it looks like those hoping for the feature on the **Galaxy S23 series** might be disappointed.

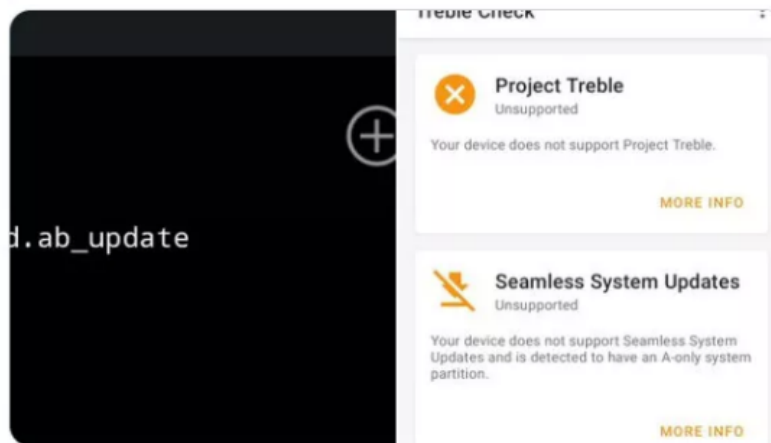
Computer Base journalist Nicolas La Rocco has **confirmed on Twitter** that the Galaxy S23 Ultra in particular doesn't support Seamless Updates. Check out the tweet below.



Nicolas La Rocco
@roccetry

Twitter/Nicolas La Rocco

The **#GalaxyS23Ultra** still doesn't support A/B partitions.



Mishaal Rahman @MishaalRahman · Feb 1

Can someone who gets their hands on an S23 today check if it uses virtual A/B partitions? twitter.com/MishaalRahman/...

[Show this thread](#)

<https://www.androidauthority.com/samsung-galaxy-s23-series-seamless-updates-3276559/>

2/5

28/04/2023, 18:35

The Samsung Galaxy S23 series doesn't support Seamless Updates

Seamless Updates were introduced back in 2016, offering a more convenient way to install system updates. The conventional method of installing a system update (and the one followed by Samsung) means users have to reboot their phone when an update is downloaded and then watch a lengthy "installing update" screen.

• • •

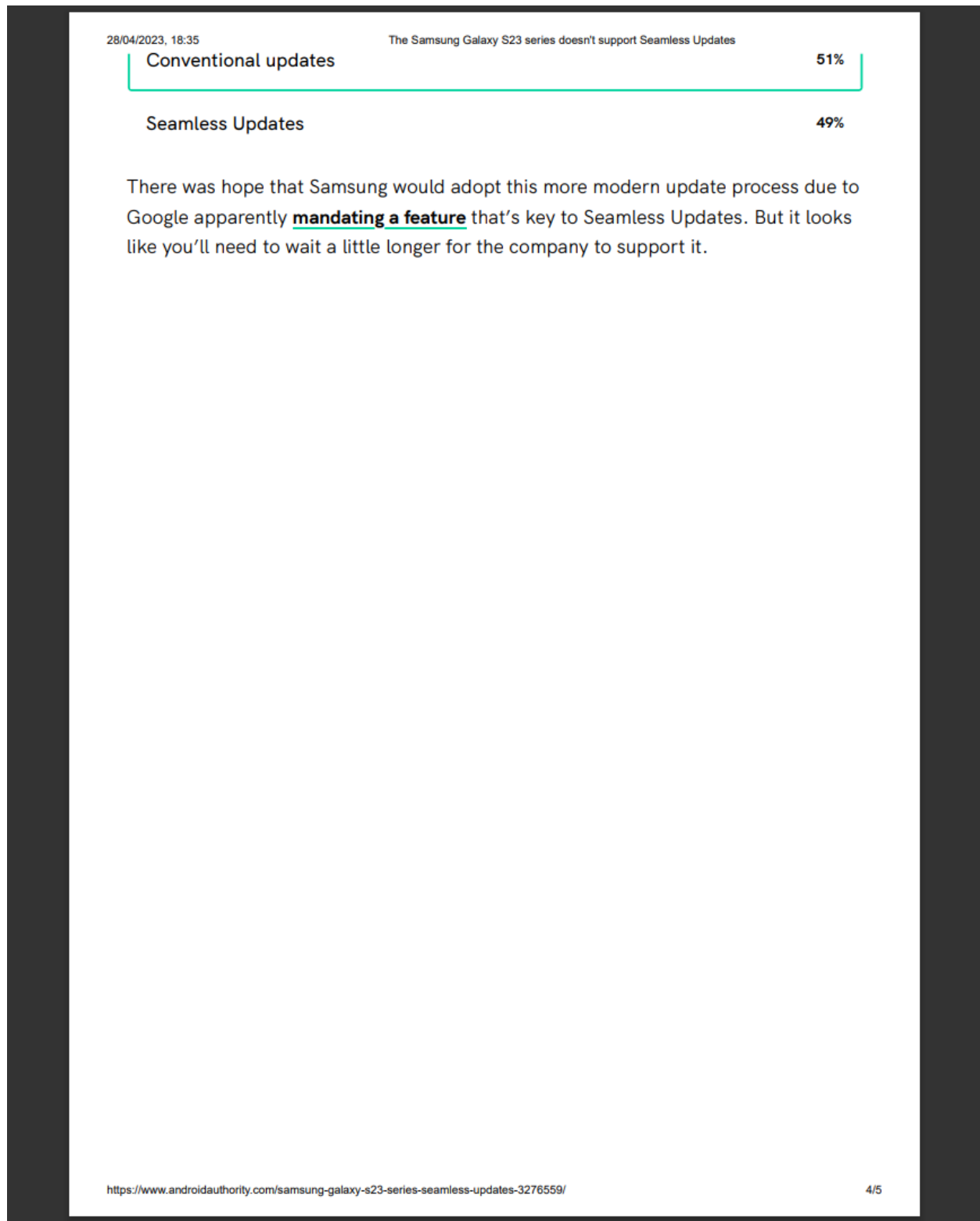
Meanwhile, Seamless Updates see the update downloaded and installed in the background on an inactive partition, while an active partition is still running so people can keep using their phone. Users then need to reboot their phone, with the device seamlessly booting into the new update by switching to the previously inactive partition. This update method requires more storage space than the conventional update process though, owing to the use of two partitions instead of one.

Do you prefer conventional updates or Seamless Updates?

1342 votes

<https://www.androidauthority.com/samsung-galaxy-s23-series-seamless-updates-3276559/>

3/5



Appendix 5 - Physical Threat Risk Report Article 1

28/04/2023, 19:22

Does The Galaxy S23 Have Face ID? All The Ways You Can Unlock The Phone

≡ 🔍

SCREEN RANT

Newsletter

Home > Tech > Tech Features

Does The Galaxy S23 Have Face ID? All The Ways You Can Unlock The Phone

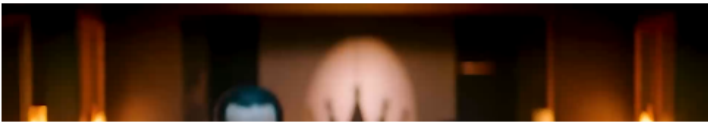
Face ID is Apple's choice of biometric authentication on iPhones, but is the feature also available on Samsung's Galaxy S23 series phones?

BY KISHALAYA KUNDU PUBLISHED FEB 12, 2023



Samsung's Galaxy S23 series comes with plenty of features for users to play around with, but does that also include Face ID? Facial recognition technology was popularized by Apple when it launched the iPhone X with Face ID in 2017. Since then, all subsequent flagship iPhones have shipped with facial recognition technology. Face unlock has also been available as a native Android feature for years, but it's not as secure as Apple's Face ID.

Like previous Galaxy S-series models, all devices in the Galaxy S23 lineup come with facial recognition technology that allows users to unlock them by just pointing the screen at their face. While many other Android vendors also offer a face unlock option in their smartphones and tablets, the underlying technology used in iPhones and Android devices is vastly different, and there are some restrictions on what can and cannot be done with it in the Galaxy S23.



<https://screenrant.com/galaxy-s23-face-unlock-options/>

1/4

28/04/2023, 19:22

Does The Galaxy S23 Have Face ID? All The Ways You Can Unlock The Phone

Link copied to clipboard





Face Unlock Limitations On The Galaxy S23

The Galaxy S23, S23 Plus, and S23 Ultra, showing their rear cameras

Image: Samsung

To use the facial recognition feature in a Galaxy S23 series device, users will first need to set a PIN, Pattern, or Password. Once that's done, go to 'Settings,' tap 'Security and privacy,' and then select 'Biometrics.' Now simply tap 'Face recognition' and then enter the PIN, password, or pattern to confirm identity. Once that's done, hit 'Continue' and then hold the phone 8-20 inches away in a way that the face is positioned inside the on-screen circle. Hold the position until the progress bar reaches 100 percent and follow on-screen instructions to register the face on the device. Once the face is registered, users can unlock their phone just by looking at it.

While face unlock offers convenience, there are a few things that users need to be aware of. Unlike Face ID in iPhones that uses dedicated hardware like a dot projector, a flood illuminator, and an infrared camera, the facial recognition technology in the Galaxy S23 (and most Android phones) only uses the front-facing camera. In addition to facial recognition, all Galaxy S23 series devices also have an ultrasonic in-display fingerprint scanner that uses specialized hardware placed underneath the screen to identify the legitimate owner, enabling them to unlock the device, access compatible apps, authorize transactions, and more.

Ad

<https://screenrant.com/galaxy-s23-face-unlock-options/>

2/4

28/04/2023, 19:22

Does The Galaxy S23 Have Face ID? All The Ways You Can Unlock The Phone

Link copied to clipboard

The lack of dedicated face unlock hardware makes the facial recognition technology in the Galaxy S23 series devices relatively less secure, meaning it can only be used to unlock the device and not to access banking apps and other high-security features. To approve payments in apps or do most other things that require biometric authentication, users will still have to use the fingerprint scanner in the **Galaxy S23**, as it's far more secure than the software-based face recognition.

MORE:

Samsung Galaxy S23 Battery Life & Charging Speeds, Explained

Source: [Samsung](#)

 Subscribe to our newsletter



Related Topics

TECH SAMSUNG

About The Author

Kishalaya Kundu
(1508 Articles Published)

Kishalaya has almost a decade's worth of experience in tech journalism, having written thousands of news, guides, features, and reviews for multiple American, Canadian, and Indian blogs. As a senior writer at ScreenRant, he covers...

[Write For Us](#) [Home](#) [Contact Us](#) [Terms](#) [Privacy](#) [Copyright](#) [About Us](#) [Press Kit](#) [Fact Checking Policy](#)
[Corrections Policy](#) [Ethics Policy](#) [Ownership Policy](#)

Copyright © 2023 screenrant.com



<https://screenrant.com/galaxy-s23-face-unlock-options/>

3/4

Appendix 6 - Physical Threat Risk Report Article 2

28/04/2023, 19:27

Galaxy S23 fingerprint scanner or biometric unlock inconsistent

PiunikaWeb

Samsung Galaxy S23 fingerprint scanner or biometric unlock inconsistent or slow for some users (potential workarounds)

Kanav Sood

Mar 27, 2023

Bugs and Issues, News, Samsung, Standalone



There's no denying that the Samsung Galaxy S23 series of smartphones are popular. However, like any popular smartphone, the S23 series mobile devices are also prone to various bugs and issues.

If you are facing difficulties in using the fingerprint scanner or biometric unlock, then you are not alone.

<https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

1/8

28/04/2023, 19:27

Galaxy S23 fingerprint scanner or biometric unlock inconsistent

**Subscribe to PiunikaWeb!**

Get updates on the latest posts and more from PiunikaWeb straight to your inbox.

SUBSCRIBE☐ I consent to receiving emails and personalized ads.

Samsung Galaxy S23 fingerprint scanner or biometric unlock inconsistent or slow

Some Samsung Galaxy S23 users (1,2,3,4,5,6,7,8,9,10), allege that the fingerprint scanner or the biometric unlock is **inconsistent**. Users also say that the phone's sensors take a lot of time to **read their fingerprints**.

And understandably, smartphone owners are only able to unlock their phones after multiple attempts.

<https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

2/8

28/04/2023, 19:27

Galaxy S23 fingerprint scanner or biometric unlock inconsistent

A Samsung Galaxy S23 owner says that even though they can use the sensor properly without their screen protector, the problem starts while it's on. Also, they are unable to manually adjust the sensor's sensitivity.

Another smartphone user mentions that the fingerprint sensor on their new Galaxy S23 Ultra is not as effective as on the old mobile devices. They also add that it can barely detect their finger.



Source

@SamsungMobile you seriously aren't able to make the fingerprint scanner work on S23? Shame! Like seriously. I have no words for this disappointing experience.

Source

Why do the biometrics work so much worse on my new galaxy 23 vs my 4 year old S10?

Source

<https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

3/8

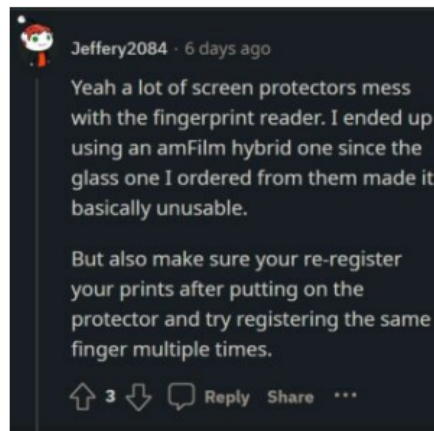
28/04/2023, 19:27

Galaxy S23 fingerprint scanner or biometric unlock inconsistent

Those affected are now requesting Samsung to fix this annoying issue as soon as possible.

It is worth mentioning that some users on the web forums speculate(1,2,3,4,5,6) the main cause of this issue to be the screen protector applied on their devices.

It appears that the glass protector adversely affects the fingerprint sensor's accuracy, let It be a new or an old one.



Source

Potential workaround

Fortunately, we did come across a workaround that may help you get rid of the problem. Firstly, you should try registering your fingerprints after **moisturizing your hands**.

Alternatively, you can try re-registering your fingerprints with the screen protector on.

<https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

4/8

28/04/2023, 19:27

Galaxy S23 fingerprint scanner or biometric unlock inconsistent



Source

Having said that, we'll keep tabs on the issue where the fingerprint scanner or biometric unlock is inconsistent or slow on Samsung Galaxy S23 series smartphones.

We will update the article as soon as we receive any new information.

Also, we have a dedicated [Samsung Galaxy S23 series updates, bugs, problems, and issues](#) tracker, so be sure to check it out.

Featured image source: [Samsung](#).

[PiunikaWeb started as purely an investigative tech journalism website with main focus on 'breaking' or 'exclusive' news. In no time, our stories got picked up by the likes of Forbes, Foxnews, Gizmodo, TechCrunch, Engadget, The Verge, Macrumors, and many others. Want to know more about us? Head here.](#)

<https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

5/8

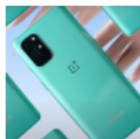
28/04/2023, 19:27

Galaxy S23 fingerprint scanner or biometric unlock inconsistent

Want to work for PiunikaWeb and enjoy best-in-industry compensation & benefits? You'll be glad to know **we're hiring experienced candidates.**

Tags : Galaxy S23 fingerprint biometric unlock inconsistent, Galaxy S23 fingerprint biometric unlock slow, Galaxy S23 fingerprint scanner inconsistent, Galaxy S23 fingerprint scanner slow, Samsung Galaxy S23 biometric inconsistent, Samsung Galaxy S23 biometric slow, Samsung Galaxy S23 fingerprint scanner inconsistent, Samsung Galaxy S23 fingerprint scanner slow

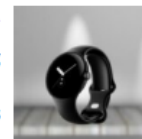
Share : [f Facebook](#) [t Twitter](#) [p Pinterest](#) [in LinkedIn](#) [✉ Email](#)

[← Previous article](#)

[Updated] Unable to use YouTube at 120Hz on your OnePlus 8T? Try these workarounds

[Next article →](#)

Google Pixel Watch charging notification delayed after March update for some users



Kanav Sood

<https://piunikaweb.com/2023/03/27/galaxy-s23-fingerprint-scanner-or-biometric-unlock-inconsistent/>

6/8